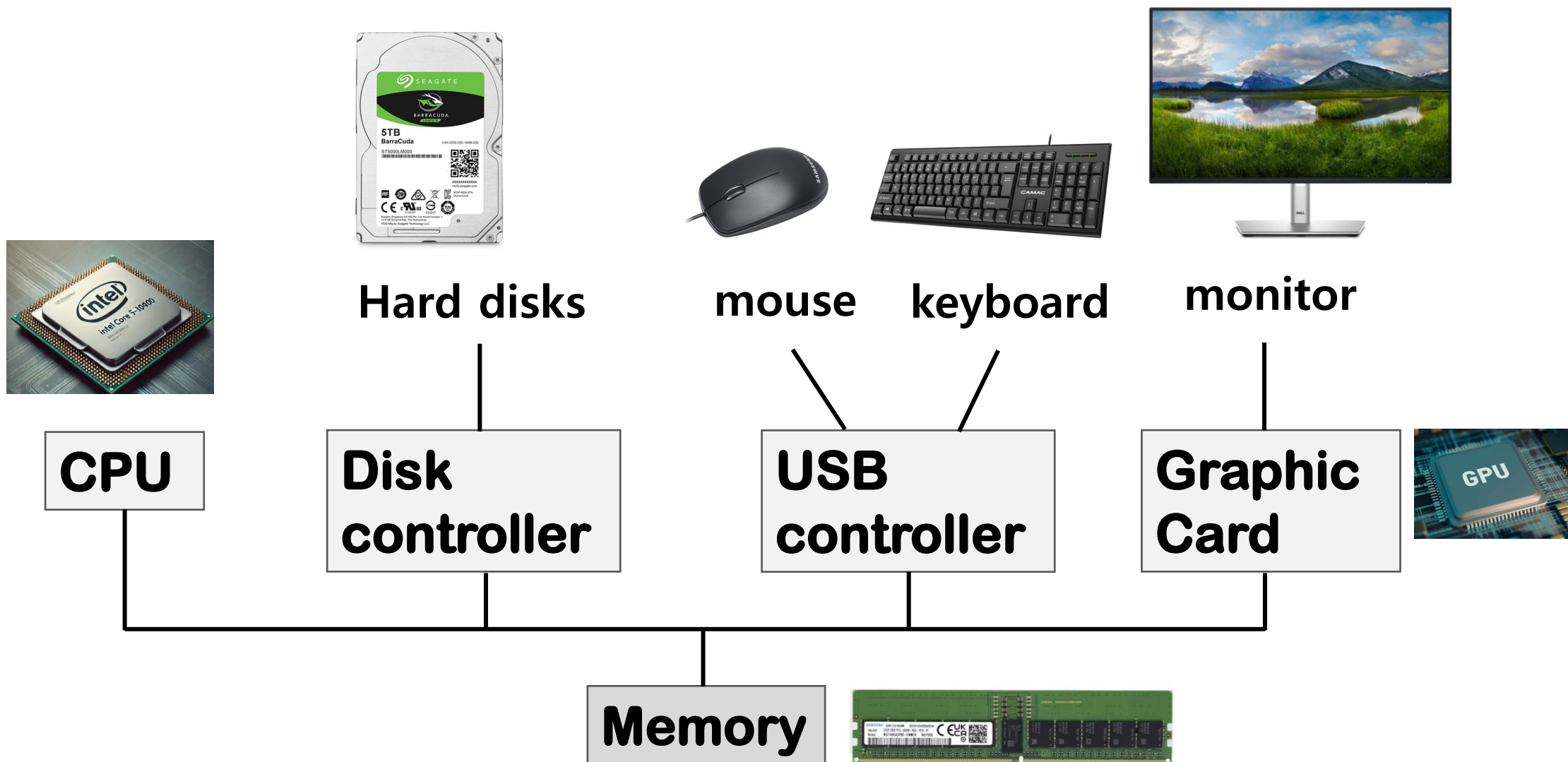


Fundamentals of Deep Learning and Programming

Lecture 1: Introduction to Python Programming

Fall 2026

Adjunct Professor: Donghoon Kang



Mainboard



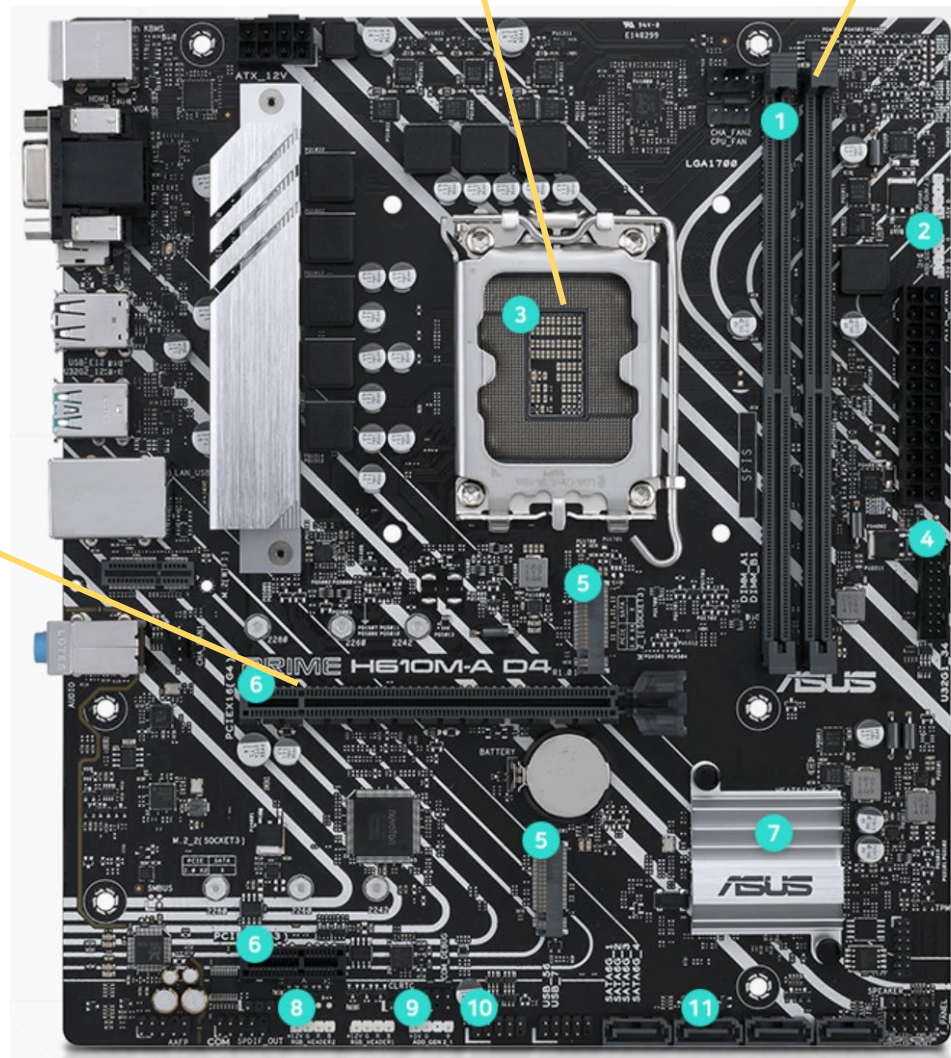
CPU



memory



Slot for Graphic card



Developing deep learning software on a **laptop** is **not recommended**.



Limited GPU performance

Insufficient cooling

Python



Version: 3.10

Advantages

- **Easy** to learn,
- suitable for developing **deep learning SW**, **web** applications

Disadvantages

- **Slow** execution time
- **High memory** usage -> **unsuitable** for developing mobile applications

Setting up Your Deep Learning Environments

For Windows



1. Install the NVIDIA GPU Driver
2. Install **Anaconda** (includes Python)
3. Install **Visual Studio Code (=VS Code)**
4. Create **Virtual Environments** in Anaconda
 - 1) Install **Pytorch** (that **includes CUDA**)
 - 2) Connect Virtual Environments in Anaconda to VS Code



Setting up Your Deep Learning Environments

For 

1. Install the NVIDIA GPU Driver

2. Install **Anaconda** (includes Python)



3. Install **CUDA Toolkit**

4. Install **Visual Studio Code (=VS Code)**



5. Create **Virtual Environments** in Anaconda

1) Install **Pytorch** (that includes CUDA)



2) Connect Virtual Environments in Anaconda to VS Code

Virtual Environments using Anaconda

Example



Virtual Environment A

Python 3.10

CUDA 11.8

Pytorch 2.0

Virtual Environment B

Python 3.11

CUDA 12.6

Tensorflow 2.21

- Isolation
- Project-specific configuration

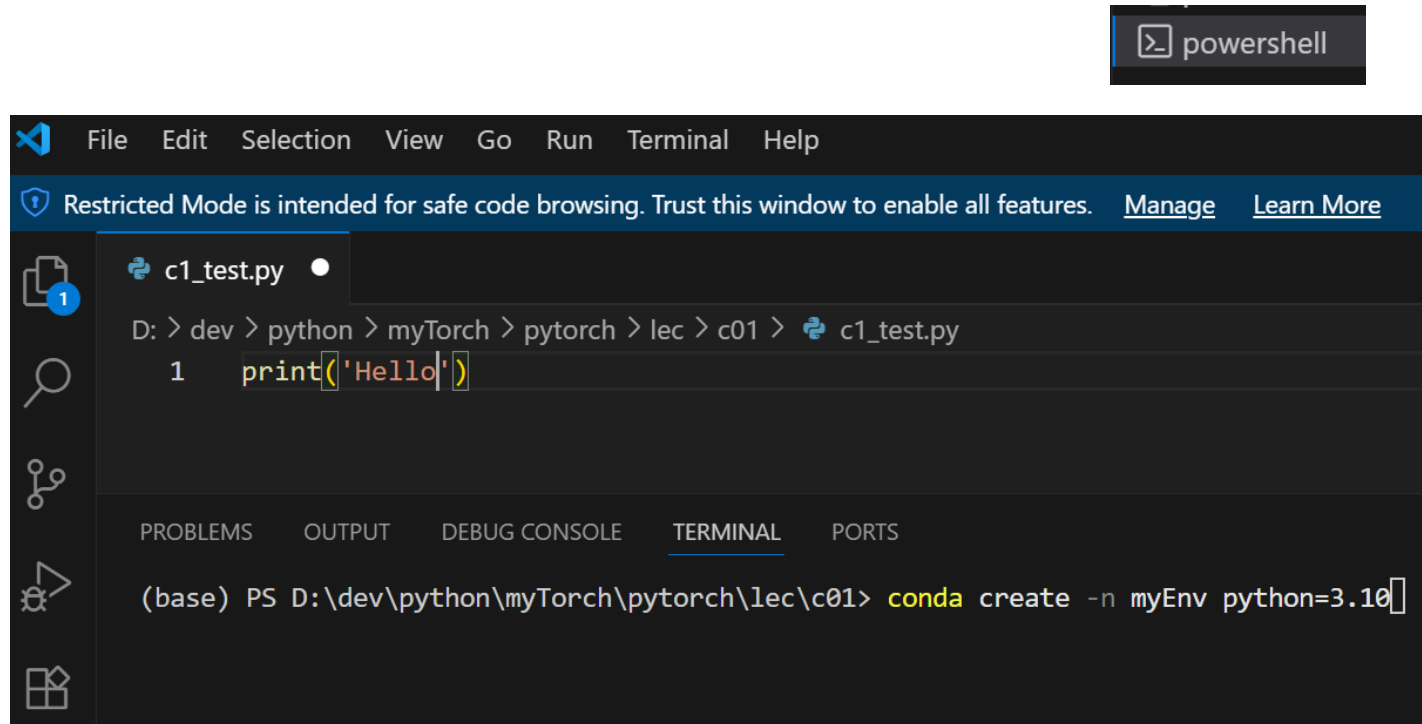
Virtual Environments using Anaconda

1. Create Virtual Environment

```
conda create -n myEnv python=3.10
```

2. Delete the Virtual Environment

```
conda remove --name myEnv --all
```



The screenshot shows a Visual Studio Code editor window. The top bar indicates 'Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage Learn More'. The editor is open to a file named 'c1_test.py' with the following content:

```
D: > dev > python > myTorch > pytorch > lec > c01 > c1_test.py
1 print('Hello')
```

The bottom panel shows the 'TERMINAL' tab with the following command prompt:

```
(base) PS D:\dev\python\myTorch\pytorch\lec\c01> conda create -n myEnv python=3.10
```

Virtual Environments using Anaconda

3. **Activate** Virtual Environment

```
conda activate myEnv
```

```
(base) PS D:\dev\python\myTorch\pytorch\lec\c01> conda activate myEnv  
(myEnv) PS D:\dev\python\myTorch\pytorch\lec\c01> 
```

4. **Deactivate** Virtual Environment

```
conda deactivate
```

```
(myEnv) PS D:\dev\python\myTorch\pytorch\lec\c01> conda deactivate  
(base) PS D:\dev\python\myTorch\pytorch\lec\c01> 
```

5. List All Virtual Environments

```
conda env list
```

```
(base) PS D:\dev\python\myTorch\pytorch\lec\c01> conda env list  
  
# conda environments:  
#  
base                * C:\Users\mo\anaconda3  
myDataAnalysis      C:\Users\mo\anaconda3\envs\myDataAnalysis  
myEnv                C:\Users\mo\anaconda3\envs\myEnv  
myEnvPytorch         C:\Users\mo\anaconda3\envs\myEnvPytorch  
myEnvTensorflow      C:\Users\mo\anaconda3\envs\myEnvTensorflow  
myPytorch            C:\Users\mo\anaconda3\envs\myPytorch
```



Python

Beginner level skills

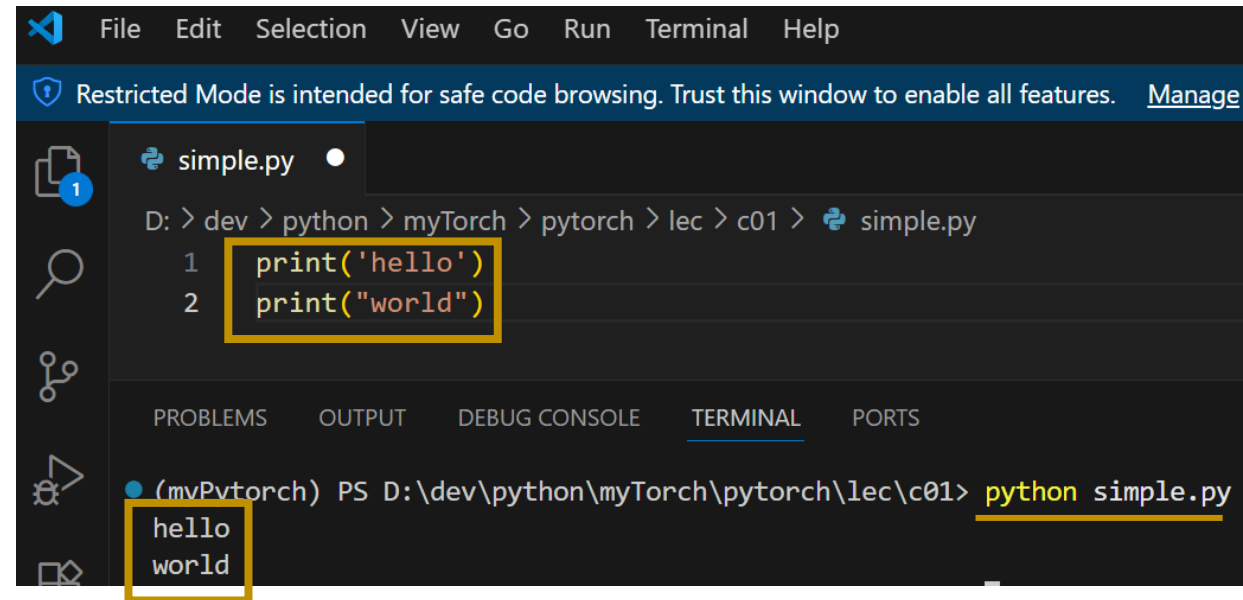
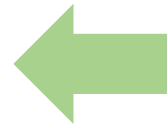
print() function



```
print('hello')  
print("world")
```



```
hello  
world
```



```
print('hello', end = ' ')  
print("world")
```



```
hello world
```



```
print('hello', end = ' test ')  
print("world")
```



```
hello test world
```



```
print('hello', end = '\n')  
print("world")
```



```
hello  
world
```

Built-in **Object Types** in Python

- numbers : -5.23 # float
- string : 'egg'
- bool : a == b # True
- **list** : [1, 3, -4]
- **tuple** : (1, 3, -4)
- set : {1, 'hello'}
- dictionary: {'age': 25}





Numbers



```
a = 1      # int
b = -5.2   # float
print(type(a))
print(type(b))
```



```
<class 'int'>
<class 'float'>
```

Numbers

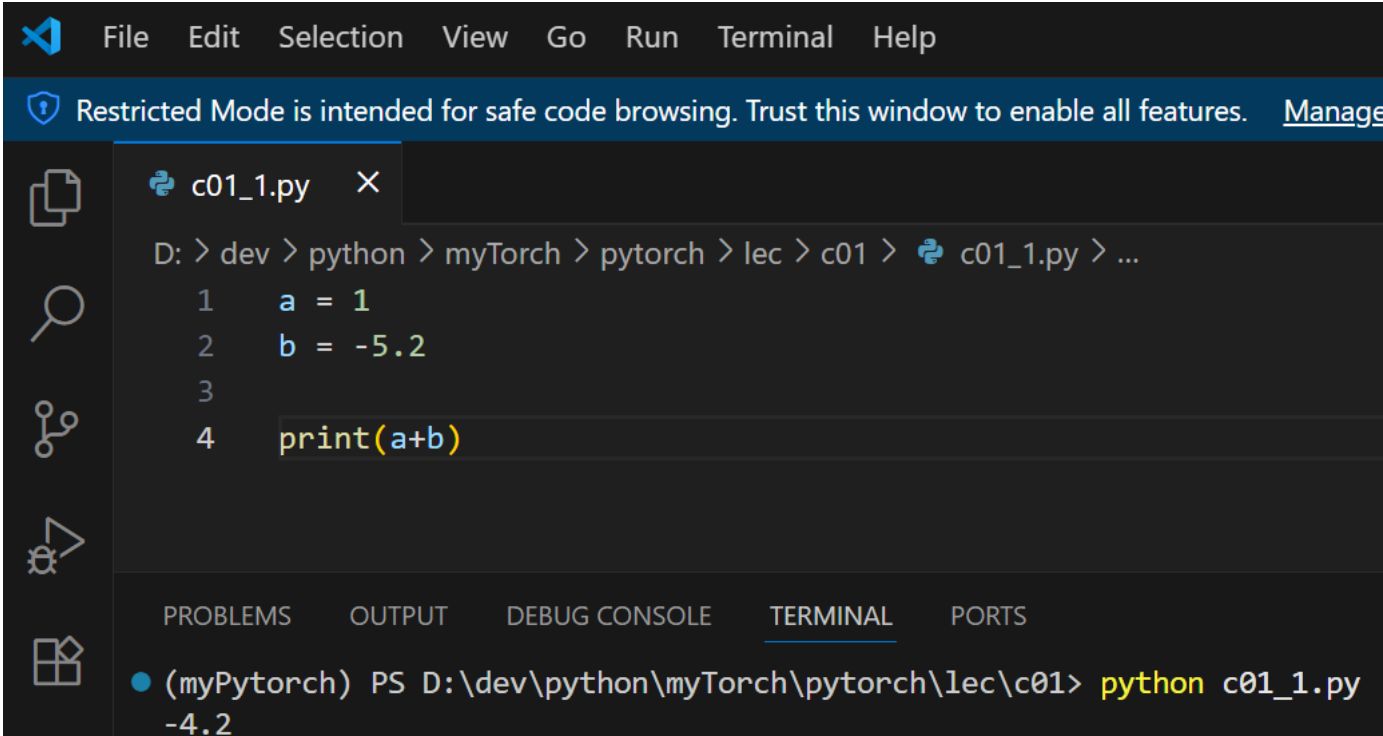


```
a = 1      # int
b = -5.2    # float

print(a)
print(b)
print(a+b) # float
```



```
1
-5.2
-4.2
```



File Edit Selection View Go Run Terminal Help

Restricted Mode is intended for safe code browsing. Trust this window to enable all features. [Manage](#)

c01_1.py ×

D: > dev > python > myTorch > pytorch > lec > c01 > c01_1.py > ...

```
1 a = 1
2 b = -5.2
3
4 print(a+b)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(myPyTorch) PS D:\dev\python\myTorch\pytorch\lec\c01> python c01_1.py
-4.2



```
a = 1
  b = -5.2 # error
```



```
b = -5.2
IndentationError: unexpected indent
```

Numbers

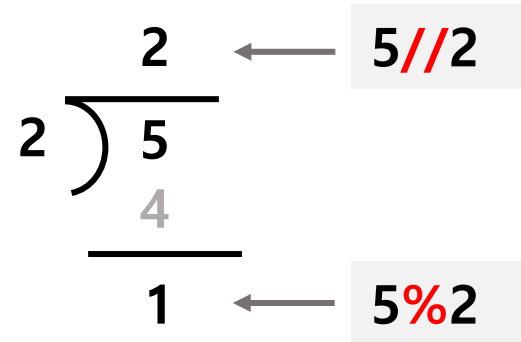


```
print(5*2)  #10
print(5/2)  #2.5
print(5//2) #2
print(5%2)  #1

Print(5**2) #52
```



```
10
2.5
2
1
25
```



Numbers



```
a = 5

a += 2    # a = a + 2
a -= 2    # a = a - 2
a *= 2    # a = a * 2
a /= 2    # a = a / 2
a //= 2   # a = a // 2
a %= 2    # a = a % 2
a **= 2   # a = a ** 2
```

`a = 1` # we **assign** a **variable** `a` to an **object** (its value is 1)

variable

a

memory address

1984297107696



object

1



```
a = 1
print(id(a))
```



1984297107696

memory address

In Python, **everything that exists in memory** is an "**object**".



String Type

```
a = 'cat'  
b = "apple"
```

String



```
a = 'Hello '  
b = "world "  
  
print(a)  
  
print(a + 3) # error
```



Hello



```
print(a+3)  
TypeError: can only concatenate str (not "int") to str
```



```
a = 'cat'  
print(len(a))
```



3



```
print(a + b)  
print(b * 3)
```



Hello world
world world world

String

indexing

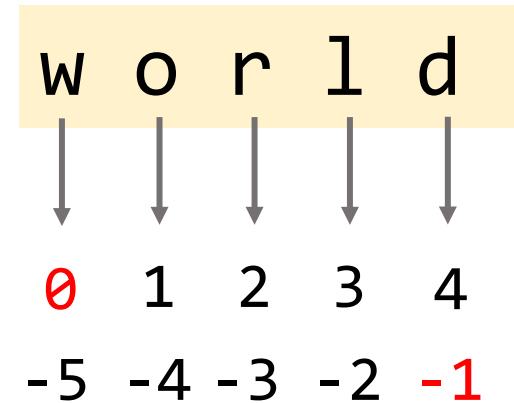


```
b = "world "
print(b[0])    #w
print(b[-1])   #d
print(b[-3])   #r
```



w
d
r

index of b



Slicing (=extracting sequences)



```
b[:4] # b[0], b[1], b[2], b[3]: slice of a from starting index 0 to 3
a[2:] # a[2], a[3], ... slice of a starting index 2 to the end
a[:]  # a == a[:]
```

start

end

Slicing (=extracting sequences)



```
a = 'world'
print(a[1:3]) # or
print(a[:3])  # wor
print(a[2:])  # rld
print(a[:])   # world
print(a[-2:]) # ld
```

~~a[1], a[2], a[3]~~

a[1:3] #slice of a from index 1 to 2

starting at
index 1

ending before index 3 (**not 3**)

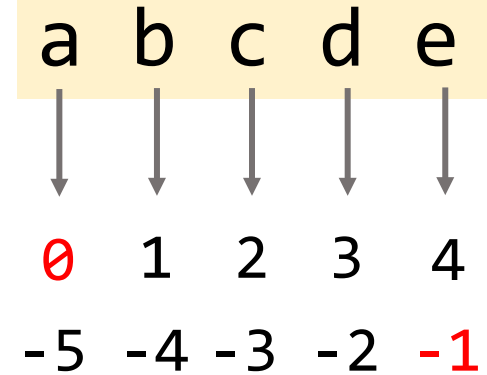
w	o	r	l	d
↓	↓	↓	↓	↓
0	1	2	3	4
-5	-4	-3	-2	-1

Slicing (=extracting sequences)



```
a = 'abcde'
print(a[1:4:1]) # == a[1:4]
print(a[1:4:2]) # == a[1],a[3]
print(a[1::2]) # a[1],a[3]
print(a[::-1]) # a[-1],..a[-5]
```

```
bcd
bd
bd
edcba
```



`a[start:end+1:step]`

String

```
a = 'Hello '
b = "world"

print(a)
print(a + b)
```

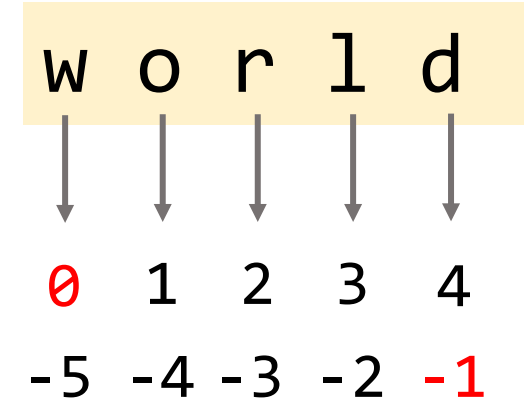
Hello
Hello world

```
a[0] = 'd' # cause error
```

```
a = 'd' + a[1:] # 'dello'
print(a)
```

dello

index of b



```
a[0] = 'd'
TypeError: 'str' object does not support item assignment
```

Str is **immutable**, meaning you can't change a string's content.

But this is OK.

`a = 'd' + a[1:]` does **not** modify the original string.

Instead, it creates a new string and reassign the variable `a` to refer to this new string



bool Type



```
a = True
b = False

c = 8 > 3 # True
d = 8 < 3 # False

print(c)
print(d)
```



```
True
False
```

bool



```
print(8 > 3)
print(8 < 3)
print(8 == 3)
print(8 != 3)
print(5 >= 5)
print(8 <= 3)
```



```
True
False
False
True
True
False
```



```
a = True or False
b = True and False
c = not (8<5) # not False
```



```
True
False
True
```

```
print(a)
print(b)
print(c)
```



List



```
[1, 3, 5]  
['cat', 3.4]  
['dog', [-2, 7]]
```

Tuple

```
(1, 3, 5)  
( 'cat', 3.4)  
( 'dog', [-2, 7])
```

List



```
a = [1, 2, 3]  
b = [4, 5, 6]
```

```
print(a + b) # concatenate a and b
```



```
[1, 2, 3, 4, 5, 6]
```



```
print(a*2)
```



```
[1, 2, 3, 1, 2, 3]
```



```
len(a) # 3
```

Tuple



```
e = (1, 2, 3)  
f = (4, 5, 6)
```

```
print(e + f) # concatenate e and f
```



```
(1, 2, 3, 4, 5, 6)
```



```
print(e*2)
```



```
(1, 2, 3, 1, 2, 3)
```



```
len(e) # 3
```

List



```
a = [1, 3, 5]
b = ['cat', 3.4]
c = ['dog', [-2, 7]]
d = []
```

```
print(a[0],a[1],a[-1])
print(b[0])
```



```
1 3 5
cat
```

```
len(a)    # 3
```

```
print(b + c)
```



```
['cat', 3.4, 'dog', [-2, 7]]
```

Tuple



```
e = (1, 3, 5)
f = ('cat', 3.4)
g = ('dog', [-2, 7])
h = ()
```

```
print(e[0],e[1],e[-1])
print(f[0])
```



```
1 3 5
cat
```

```
len(e)    # 3
```

```
print(f + g)
```



```
('cat', 3.4, 'dog', [-2, 7])
```

List



```
a = list(1,3,5) # error
```



```
a = list(1,3,5)
TypeError: list expected at most 1 argument, got 3
```



```
a = list([1,3,5])
b = list((1,3,5))
c = list('cat')
d = list(range(1,4))
```

```
[1, 3, 5]
[1, 3, 5]
['c', 'a', 't']
[1, 2, 3]
```

```
print(a)
print(b)
print(c)
print(d)
```

Tuple



```
a = tuple(1,3,5) # error
```



```
e = tuple(1, 3, 5)
TypeError: tuple expected at most 1 argument, got 3
```



```
e = tuple([1,3,5])
f = tuple((1,3,5))
g = tuple('cat')
h = tuple(range(1,4))
```

```
(1, 3, 5)
(1, 3, 5)
('c', 'a', 't')
(1, 2, 3)
```

```
print(e)
print(f)
print(g)
print(h)
```

Mutable vs. Immutable

IMPORTANT

List

```
a = [1, 3, 5]
a[0] = 9 # OK. List is a mutable (=changeable) type
print(a)
```



9 3 5

Tuple



```
e = (1, 3, 5)
e[0] = 9 # error 
# Tuple is an immutable (=unchangeable) type
```

```
e[0] = 9
TypeError: 'tuple' object does not support item assignment
```

Slicing

List

```
a = [1, 3, 5, 7]
print(a[1:2])
print(a[:2])
print(a[1:])
print(a[:])
```



```
[3]
[1, 3]
[3, 5, 7]
[1, 3, 5, 7]
```

Tuple

```
b = (1, 3, 5, 7)
print(b[1:2])
print(b[:2])
print(b[1:])
print(b[:])
```



```
(3)
(1, 3)
(3, 5, 7)
(1, 3, 5, 7)
```

a = [1, 3, 5, 7]

Index of a

0 1 2 3
-4 -3 -2 -1

Nesting

List

```
a = [[1, 2, 3],  
      [-2, -3, 10]]
```

```
print(a[1])  
print(a[1][2])
```

```
</> [-2, -3, 10]  
10
```

```
a[1][2] = -5  
print(a[1][2])
```

```
</> -5
```

Tuple

```
b = ((1, 2, 3),  
      (-2, -3, 10))
```

```
print(b[1])  
print(b[1][2])
```

```
</> (-2, -3, 10)  
10
```

```
b[1][2] = -5 # error  
print(b[1][2])
```

tuple is an **immutable** type

```
b[1][2] = -5  
TypeError: 'tuple' object does not support item assignment
```

Tuple



```
a = (1, 3, 5) # tuple  
b = 1, 3, 5  # tuple
```

```
c = 1      # int  
d = 1,     # tuple  
e = (1,)   # tuple  
f = (1)    # int  
print(type(d))  
print(type(e))  
print(type(f))ZZZZZ
```



```
<class 'tuple'>  
<class 'tuple'>  
<class 'int'>
```

```
g = (1 2) # error
```



```
g = (1 2) # cause error  
    ^^^
```

```
SyntaxError: invalid syntax. Perhaps you forgot a comma?
```

Assignment of multiple variables



```

a = 3
b = 5
c, d = (-2, 7) # c, d = -2, 7
e, f = [1, 9]

print(a, b)
print(c, d)
print(e, f)

```



```

3 5
-2 7
1 9

```



```

a = 3
b = 5
print(a, b)
a, b = b, a
print(a, b)

```



```

3 5
5 3

```



Set Type

- Unordered
- Unique
- Mutable

type



```
a = {1, 'hello'}  
b = set([1, 'hello'])
```

```
c = {1, -5, 8, 8}  
print(c)
```



```
{8, 1, -5}
```

set



```
A = {1, 2, 3}
```

```
B = {3, 4, 5}
```

```
print(A | B)    # Union: {1, 2, 3, 4, 5}
```

```
print(A & B)    # Intersection: {3}
```

```
print(A - B)    # Difference: {1, 2}
```

```
print(A ^ B)    # Symmetric difference: {1, 2, 4, 5}
```

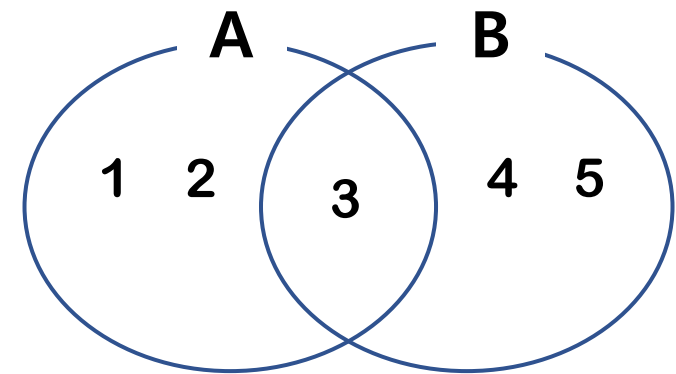


```
{1, 2, 3, 4, 5}
```

```
{3}
```

```
{1, 2}
```

```
{1, 2, 4, 5}
```





Dictionary Type



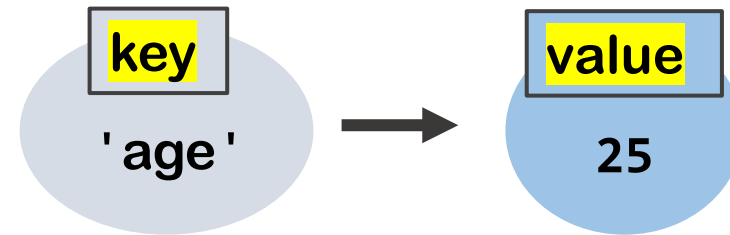
```
a = {'age': 25}
print(a)
a['age'] = 30
print(a)
```

mutable type



```
{'age': 25}
{'age': 30}
```

- “Keys” must be unique.
- “Values” can be duplicated



dictionary



```
b = {'age': 25, 'myList': [2,3,4]}
```

```
print(b)  
print(b['age'])  
print(b['myList'])  
print(type(b))
```



```
{'age': 25, 'myList': [2, 3, 4]}  
25  
[2, 3, 4]  
<class 'dict'>
```



```
c = dict(age = 25, myList = [2,3,4])  
print(c)
```



```
{'age': 25, 'myList': [2, 3, 4]}
```



if statement



```
age = 30
```

```
if age > 40:  
    print('old')  
else:  
    print('young')
```



young

while loop

```
count = 0
```

```
while count <= 5:  
    print(count)  
    count += 2
```

0
2
4

for loop

```
myList = [2, -4, 30]
```

```
for i in myList:  
    print(i)
```

2
-4
30

if statement



```
score = 85

if score >= 90:
    print('A')
else:
    if score >= 80:
        print('B')
    else:
        print('C or lower')
```



B

=



```
score = 85

if score >= 90:
    print('A')
elif score >= 80:
    print('B')
else:
    print('C or lower')
```



B

if statement



```
x = 15

if x > 10:
    if x < 20:    # nested if
        print("x is between 10 and 20")

if 10 < x < 20:
    print("x is between 10 and 20")
```



```
x is between 10 and 20
x is between 10 and 20
```



```
menu = ('milk', 'egg')
choice = 'egg'

if choice in menu:
    print('OK')
else:
    print('Unavailable')
```



OK

while loop

```
count = 0
```

```
while count <= 5:  
    print(count)  
    count += 2
```



```
0  
2  
4
```

```
n = 1
```

```
while True: # infinite loop  
    print(n)  
    if n == 3:  
        break # stop the loop  
    n += 1
```



```
1  
2  
3
```

```
n = 1
```

```
while n < 5:  
    n += 1  
    if n == 3:  
        continue # skip the rest of the loop for this iteration  
    print(n)
```



```
2  
4  
5
```

while loop



```
while n < 5:  
    print(n)  
    if n == 3:  
        break # stop the loop  
    n += 1  
else:  
    print('loop finished')
```



```
1  
2  
3
```



```
while n < 5:  
    print(n)  
    n += 1  
else:  
    print('loop finished (no break)')
```



```
1  
2  
3  
4  
loop finished (no break)
```

for loop



```
myList = [2, -4, 30]
```



```
for i in myList:  
    print(i)
```

```
2  
-4  
30
```



```
a = [(1,2), (3,4), (5,6)]
```



```
for (first, last) in a:  
    print(first, last)
```

```
1 2  
3 4  
5 6
```



```
for i in range(4):  
    print(i)
```

```
0  
1  
2  
3
```



```
for i in range(2,5):  
    print(i)
```

```
2  
3  
4
```

range(..)

- `range()` is a function that generates a sequence of numbers in Python.
- `range()` is **not a list**, but a **lazy sequence object**.



```
start = 10
end = 30
step = 5
```

```
for i in range (start, end+1, step):
    print(i)
```

```
10
15
20
25
30
```



```
print(range(4))
```

```
range(0, 4)
```

```
for i in range(2,5):
    print(i)
```

```
2
3
4
```

```
print(list(range(4)))
```

```
[0, 1, 2, 3]
```

```
for i in range (8, 1, -2):
    print(i)
```

```
8
6
4
2
```

Break and continue in for loop



```
for i in range(1, 5):
    if i == 3:
        print("Breaking the loop at i =", i)
        break # completely exits the loop
    print("Current value:", i)
```



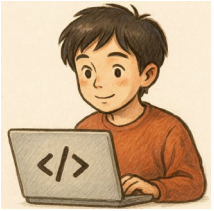
```
Current value: 1
Current value: 2
Breaking the loop at i = 3
```



```
for i in range(1, 5):
    if i == 3:
        print("Skipping this iteration at i =", i)
        continue # skip the rest of the loop
    print("Current value:", i)
```



```
Current value: 1
Current value: 2
Skipping this iteration at i = 3
Current value: 4
```



Functions



```
def myfunc():  
    print('hello')
```

```
myfunc()
```



```
hello
```



```
def add(x, y):  
    return x+y
```

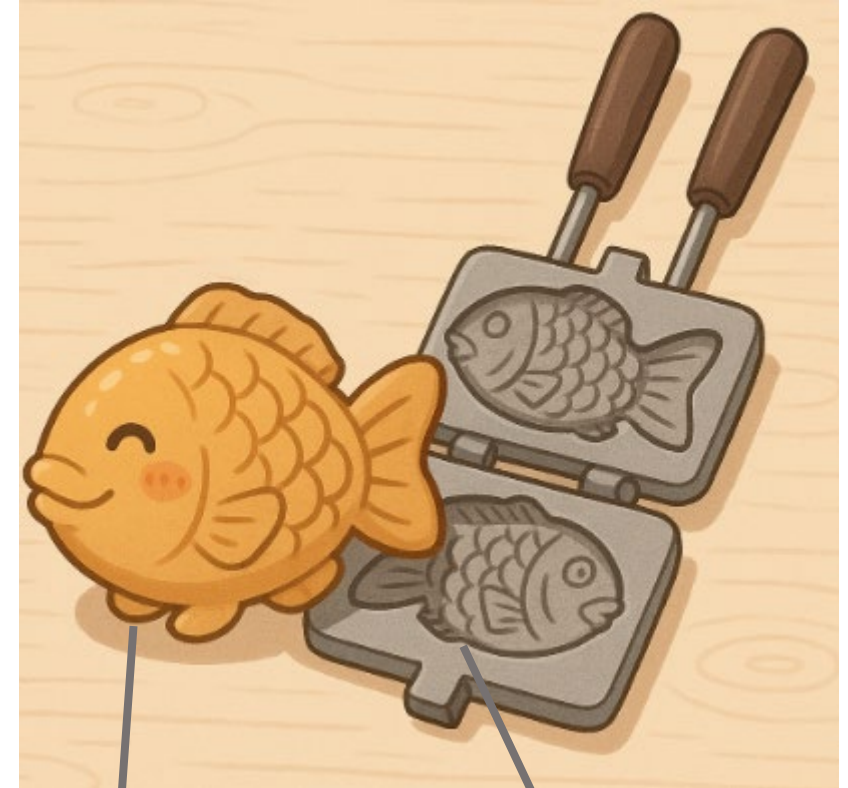
```
print(add(2,5))
```



```
7
```



Class



myFish

(**instance** of class)
(=**object**)

Fish

(class name)

```

class Fish:
    pass

myFish = Fish()
    
```

Variables and Methods in class



```
class Dog:
    def bark(self):      # method
        print('mung mung')

my_dog = Dog()
my_dog.bark()
```

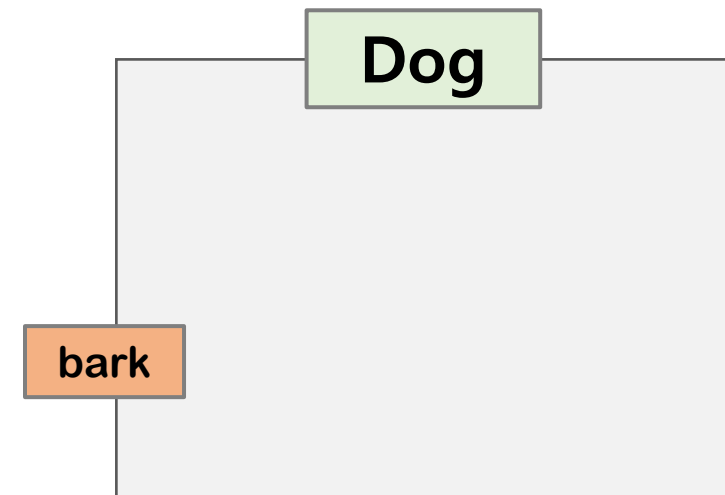
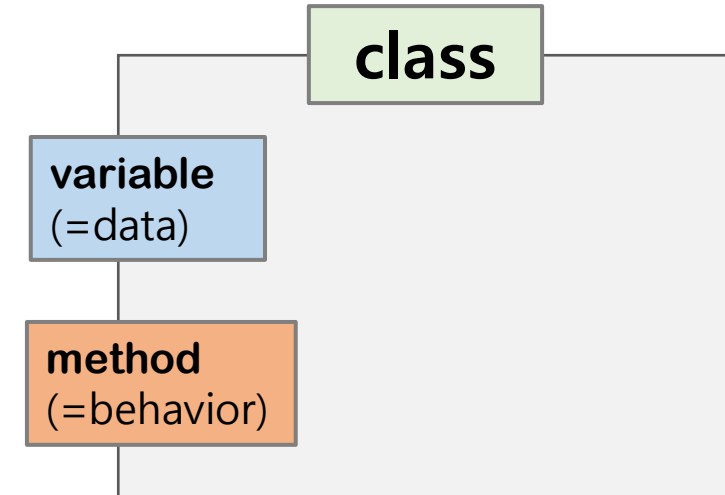


```
mung mung
```



In Python, an **object** is an **instance of a class** that contains both **variables (data)** and **methods (behaviors)**.

- **variables:** Variables that belong to the object.
- **methods:** Functions that belong to the object and can operate on its data.



`__init__(self, ..)`



```
class Dog:
    def __init__(self, dog_name): # initialize
        self.name = dog_name # instance variable

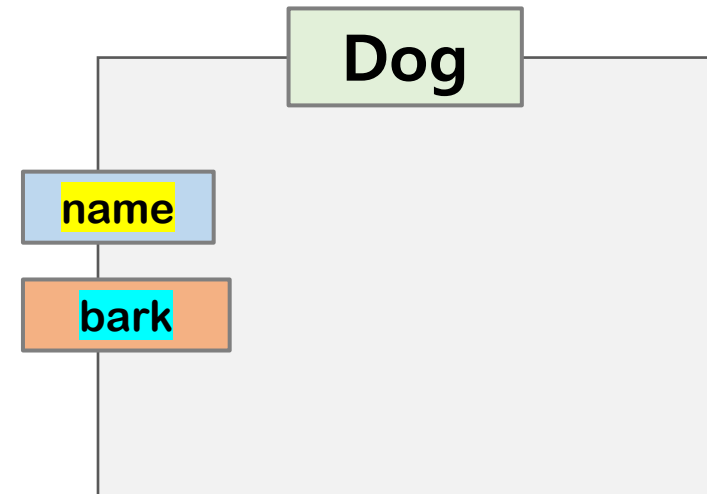
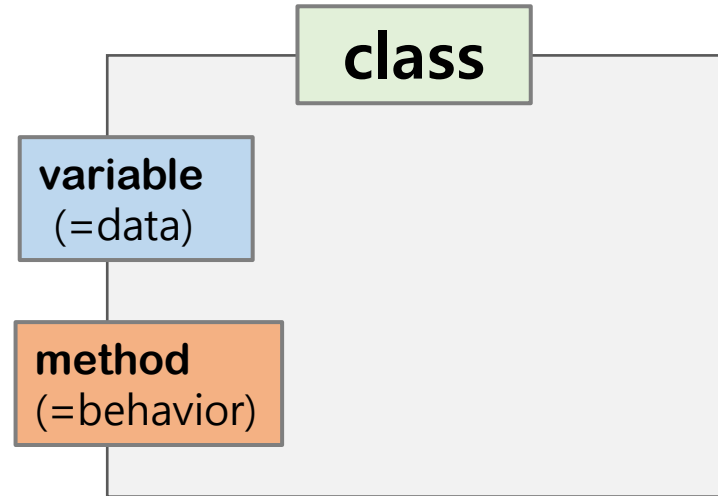
    def bark(self): # method
        print(f'{self.name}: mung mung')

my_dog = Dog('kongi')
my_dog.bark()

your_dog = Dog('baduki')
your_dog.bark()
```



```
kongi: mung mung
baduki: mung mung
```





Exceptions



```
try:
    4/0
except:
    print('error')
```



error



```
try:
    4/0
except Exception as e:
    print(e)
else:
    print(no Exception error)
finally:
    print('finally')
```



division by zero
finally



```
try:
    pass
except Exception as e:
    print(e)
else:
    print(no Exception error)
finally:
    print('finally')
```



no Exception error
finally



User Input/Output (1)

Input() function



```
a = input()  
print(a + ' world')
```



```
hello  
hello world
```

```
a = input('Please enter your name: ')  
print('Hi ' + a)
```

```
Please enter your name: chulsoo  
Hi chulsoo
```

User Input/Output (2)

Open() function (1)

The `open( )` function is used to **open a file** in Python. It **returns a file object**, which allows you to read from or write to the file.

```
open(file, mode='r', encoding=None)
```

mode: The mode in which the file is opened. Some common modes are:

'r' → Read (default). File must exist.

'w' → Write. Creates a new file or overwrites existing content.

'a' → Append. Adds data to the end of the file.

encoding: Encoding type (e.g., "utf-8") when working with text files.

Reading a file



```
file = open("example.txt", "r", encoding="utf-8")
content = file.read() # Read the whole content
print(content)

file.close() # Always close the file
```

User Input/Output (3)

Open() function (2)

Writing a file

```
file = open("example.txt", "w", encoding="utf-8")  
content = file.write("Hello world") # write some text to the file  
file.close() # Always close the file
```



```
with open("example.txt", "w", encoding="utf-8") as file:  
    content = file.write("Hello world") # write some text to the file
```

- When the program leaves the with block, file.close() is called automatically.
- This makes the code safer and cleaner.

User Input/Output (4)

Open() function (3)



Example.txt

```
Hello world  
Hi there !
```



```
with open("example.txt", "r", encoding="utf-8") as file:  
    for line in file:  
        print(line)
```



```
Hello world  
Hi there !
```



Python

Intermediate level skills



Modules



Module = a single Python file



Package = a folder that contains multiple modules

Modules and Packages



Module = a single **Python file**



Package = a **folder** that contains multiple modules

For Example

```

project/
|— main.py # module. entry point (= main script)
|— say.py # module
|
|— myPackage/ # package (folder)
    |— __init__.py # a special Python file that tells Python "this folder is a package."(< Python 3.2) and performs initialization
    |— calc.py # module
  
```

How to use functions in other modules



```
project/  
├── main.py  
├── say.py  
└── myPackage/  
    └── calc.py
```



say.py

```
def hello():  
    print('hello world')
```



myPackage



calc.py

```
def add (x, y):  
    return x + y
```



main.py

```
import say  
say.hello() # hello world
```

```
import myPackage  
print(myPackage.calc.add(2,3)) # error because there is no __init__.py
```



hello world

```
print(myPackage.calc.add(2,3))  
AttributeError: module 'myPackage' has no attribute 'calc'
```

How to use functions in other modules



```
project/  
├── main.py  
├── say.py  
└── myPackage/  
    ├── __init__.py  
    └── calc.py
```



```
say.py  
  
def hello():  
    print('hello world')
```



myPackage



```
__init__.py  
  
from . import calc
```



```
calc.py  
  
def add (x, y):  
    return x + y
```



```
main.py  
  
import say  
say.hello() # hello world  
  
import myPackage  
print(myPackage.calc.add(2,3)) # OK
```



```
hello world  
5
```

=

```
main.py  
  
import say  
say.hello() # hello world  
  
from myPackage.calc import add  
print(add(2,3)) # OK
```



```
hello world  
5
```

```
__name__ == "__main__"
```



say.py

```
def hello():  
    print('hello world')  
  
if __name__ == "__main__":  
    print("my_module.py is running directly")
```



my_module.py is running directly



world.py

```
import say  
say.hello() # hello world
```



hello world



String Formatting

f formatting (after Python 3.6)

```
name = kongi  
age = 5  
X = f'{name} is {age} years old.'  
  
print(X)
```



kongi is 5 years old.

str Formatting (1)



```
a = 'I have %d apples.' % 4  
print(a)
```



I have 4 apples.

% d # OK



```
b = 'The dog weighs %f kg.' % 5.2  
print(b)
```



The dog weighs 5.2 kg.



```
d = 'I have %d %s' % (4, 'apples.')  
print(d)
```



I have 4 apples.

str formatting Method Call (2)



```
result = '{} {} weigh {} grams.'.format(10, 'eggs', 605.2)  
print(result)
```



10 eggs weigh 605.2 grams.



```
result = '{0} {1} weigh {2} grams.'.format(10, 'eggs', 605.2)  
print(result)
```



10 eggs weigh 605.2 grams.



```
X = '{0} {food}'.format(9, food='apples')  
print(X)
```



9 apples

f str Formatting (after python 3.6)



```
name = 'kongi'  
age = 5  
X = f'{name} is {age} years old.'  
print(X)
```



```
kongi is 5 years old.
```



Methods of lists, strings, sets, and dictionaries

List Methods (1)



```
a = [1, 2, 3]
b = a
a.append(4)    # in-place operation
print(a)      # [1, 2, 3, 4]
print(b)      # [1, 2, 3, 4] -> b is also changed since it references the same object
```



```
[1, 2, 3, 4]
[1, 2, 3, 4]
```

In Python, **in-place** operation **modifies** the **existing object directly**.

- ✓ **In-place** = modifies the existing object
- ✓ **Non in-place** = creates and returns a new object

List Methods (2)



Non In-place



```
a = [1, 2]
b = a
a = a + [3] # non in-place: creates and returns a new object
print(a) # [1, 2, 3]
print(b) # [1, 2] -> variable a points to a new object
```



```
[1, 2, 3]
[1, 2]
```

In-place



```
a = [1, 2]
b = a
a += [3] # in-place
print(a) # [1, 2, 3]
print(b) # [1, 2, 3] -> the original object was modified
```



```
[1, 2, 3]
[1, 2]
```

List Methods (3)



print(num)



```
num = [1, 2, 3]
num.extend([4, 5]) # [1, 2, 3, 4, 5]
print(num)
```



[1, 2, 3, 4, 5]

```
num = [7, 8, 9, 10]
num.insert(1, 50) # [7, 50, 8, 9, 10]
```

```
num = [7, 8, 9, 10]
a = num.pop() # num == [7, 8, 9], a == 10
```

List Methods (4)



```
num = [7, 8, 9, 10]
num.remove(9) # num == [7, 8, 10]
print(num)
```



[7, 8, 10]

```
num = ['cat', 'dog', 'horse']
a = num.index('cat') # a == 0
print(a)
```



0

```
num = [1, 4, 3, 2]
num.sort() # [1, 2, 3, 4]
print(num)
```



[1, 2, 3, 4]

```
num = [7, 8, 9, 10]
num.reverse() # [10, 9, 8, 7]
print(num)
```



[10, 9, 8, 7]

del

- ✓ **del** is not a function, but a **Python keyword**.
- ✓ That means it's a language construct, not something like `print( )` or `len( )`.
- ✓ **del** can delete not only list elements but also whole variables



```
num = [7, 8, 9, 10]
del num[2:] # [7, 8]
print(num)
```



```
[7, 8]
```

```
x = 100
del x
print(x) # Error: x was deleted
```

```
print(x)
NameError: name 'x' is not defined
```

```
x = (2, 4, 5)
del x
print(x) # Error: x was deleted
```

```
print(x)
NameError: name 'x' is not defined
```

String Methods (1)



```
a = 'hello'  
print(a.find('e'))
```



1

```
a = ','.join('abcd')  
print(a) # a, b, c, d
```



a,b,c,d

```
a = 'HELLO' # string is immutable  
b = a.lower() # non in-place  
print(a)  
print(b)
```



HELLO
hello

String Methods (2)



```
a = 'hello world'
b = a.split()
print(a)
print(b)
```



```
hello world
['hello', 'world']
```

```
c = 'a:b:c:d'
d = c.split(':')
print(c)
print(d)
```



```
a:b:c:d
['a', 'b', 'c', 'd']
```

```
a = 'hello world'
b = a.replace('hello', 'hi')
print(a)
print(b)
```



```
hello world
hi world
```

Set Methods



```
s = {7, 8, 9}
s.add(50)      # {7, 8, 9, 50}
```

```
s = {7, 8, 9}
s.remove(8)    # {9, 7}
```

```
s = {7, 8, 9}
s.update([-10, 3]) # {3, 7, 8, 9, -10}
```

```
A = set([1, 2, 3, 4]) # {1, 2, 3, 4}
B = set([3, 4, 9, 20]) # {3, 4, 9, 20}
A & B      # {3, 4}
A | B      # {1, 2, 3, 4, 9, 20}
A - B      # {1, 2}
```

Dictionary Methods



```
d = {'name': 'chulsoo', 'age': 21 }  
print(d.keys())  
print(d.values())  
print(d.items())  
print(d.clear())
```



```
dict_keys(['name', 'age'])  
dict_values(['chulsoo', 21])  
dict_items([('name', 'chulsoo'), ('age', 21)])  
None
```



```
d = {'name': 'chulsoo', 'age': 21 }  
a = 'name' in d  
b = 'weight' in d  
print(a, b)
```



```
True False
```



Copy List



```
a = [1, 2, 3]
b = a

print(id(a), id(b))
print (a is b)

a[0] = 8
print(a, b)
```



```
2324562958528 2324562958528
True
[8, 2, 3] [8, 2, 3]
```



```
from copy import copy

c = [1, 2, 3]
d = c[:] # d = c.copy()

print(id(c), id(d))
print (c is d)

c[0] = 8
print(c, d)
```



```
2277206448320 2277206872320
False
[8, 2, 3] [1, 2, 3]
```

- ✓ Variables, a and b do **not** create two separate lists.
- ✓ They both **reference (or point to) the same list object** in memory, so they share the same underlying data."



Shallow Copy vs Deep Copy

(1) Shallow Copy



```
import copy

original = [[1, 2, 3], [4, 5, 6]] # Original list with nested list
shallow_copied = copy.copy(original) # Shallow copy: the same as shallow_copy = original
shallow_copied[0][0] = 99 # Modify nested list inside the shallow copy

print("Original:", original)
print("Shallow Copy:", shallow_copied)
```



```
Original: [[99, 2, 3], [4, 5, 6]]
Shallow Copy: [[99, 2, 3], [4, 5, 6]]
```

- ✓ A shallow copy creates a new object, but does not recursively copy nested objects.
- ✓ Instead, it just copies references of nested objects.
- ✓ This means that if the original object contains lists (or other mutable objects) inside, both the original and the copied object will point to the same inner objects.

Shallow Copy vs Deep Copy

(2) Deep Copy



```
import copy

original = [[1, 2, 3], [4, 5, 6]] # Original list with nested list

# Deep copy: Creates a completely independent clone, including all nested objects.
deep_copied = copy.deepcopy(original)
deep_copied[0][0] = 99 # Modify nested list inside the deep copy

print("Original:", original)
print("Deep Copy:", deep_copied)
```

```
Original: [[1, 2, 3], [4, 5, 6]]
Deep Copy: [[99, 2, 3], [4, 5, 6]]
```

- ✓ A deep copy creates a completely independent copy of the object, including all nested objects.
- ✓ This means that modifications in the deep copy will not affect the original object.



Mutable vs Immutable Object Types (1)

In Python, the difference between mutable and immutable objects mainly shows up when you **pass them into a function**.

(1) **Immutable objects** (like **int, float, str, tuple**)

cannot be **changed in place**. If you try to change them, Python creates a new object.



```
def add_one(x):  
    x = x + 1    # This creates a NEW int object  
    print("Inside function:", x)  
  
num = 10        # int: immutable object  
add_one(num)  
print("Outside function:", num)
```



```
Inside function: 11  
Outside function: 10
```

Mutable vs Immutable Object Types (2)

(2) Mutable objects (like list, dict, set)

can be changed in place, so modifications inside a function will affect the original object even after the function ends.



```
def add_item(my_list):  
    my_list.append(100)    # Modify in place  
    print("Inside function:", my_list)
```

```
nums = [1, 2, 3] # list: mutable object  
add_item(nums)  
print("Outside function:", nums)
```



```
Inside function: [1, 2, 3, 100]  
Outside function: [1, 2, 3, 100]
```

Arguments of Function (1)

(1) Positional or keyword arguments

```
def f(a, b):  
    print(a, b)  
  
f(1, 2)          # positional  
f(a=1, b=2)     # keyword
```



```
1 2  
1 2
```

(2) Arbitrary arguments

```
def f(a, *args):          #*args: arbitray  
    print(a, args)  
  
f(1, 2, 3, 4)            # a=1, args=(2,3,4)
```



```
1 (2, 3, 4)
```

The parameter `*args` collects all arguments into a tuple.

Arguments of Function (2)

(3) Order of arguments: f(normal, *arbitray, **keyword)

```
def f(x, y, *args, **kwargs):  
    print(args, x, y, kwargs)
```

```
f(1, 2, 3, 'hello', age=20)
```



```
(3, 'hello') 1 2 {'age': 20}
```





class

(1) Simplest class



```
class Dog:
    pass

my_dog = Dog()
```

(2) Initialization



```
class Dog:
    def __init__(self):
        print('initialization')

my_dog = Dog()
```



initialization



class

Variables and Methods

```
class Dog:
    def __init__(self, dog_name):
        self.name = dog_name      # instance variable
        self.__age = 5           # special variable

    def bark(self):               # method
        print(f'{self.name} ({self.__age}): mung mung')

my_dog = Dog('kongi')
my_dog.bark()
```

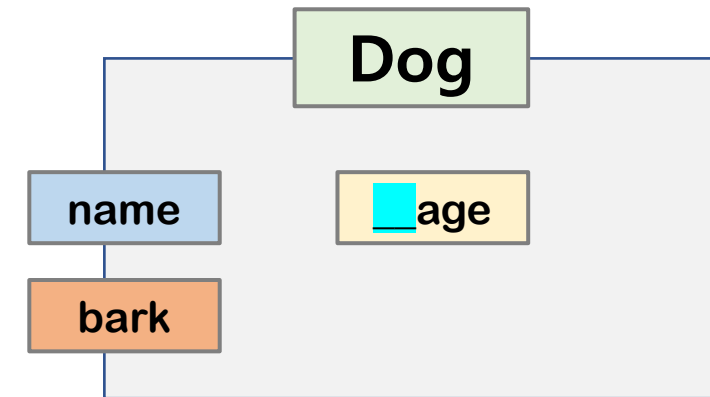
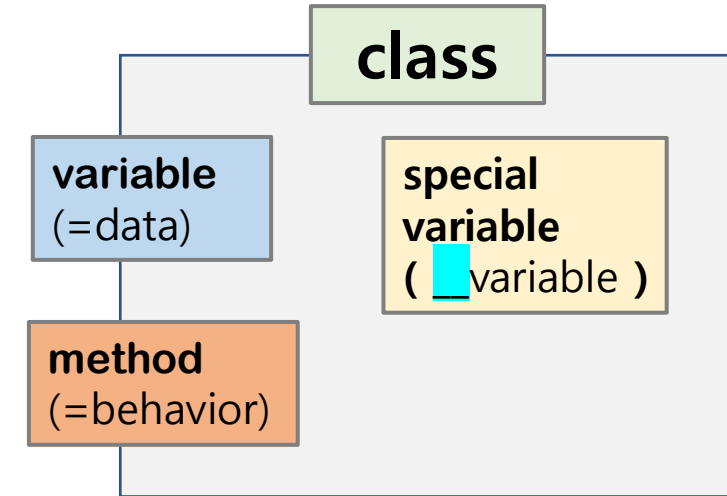


kongi (5): mung mung

```
my_dog.name = 'baduki'
my_dog.__age = 10
my_dog.bark()
```



baduki (5): mung mung



self.__age = 5 (internally stored as self._Dog__age due to name mangling)

Class Variables

```
class Dog:
    num_leg = 4                # class variable (shared class data)

    def __init__(self, dog_name):
        self.name = dog_name   # instance variable (per-instance data)

my_dog = Dog('kongi')
your_dog = Dog('baduki')

print(my_dog.num_leg)
print(Dog.num_leg)
```



4
4

Class Inheritance (1)



```
class Parent:
    pass

class Child(Parent):
    pass

my_child = Child()
```



```
class Dog:
    def bark(self): # OK
        print('mung')

class Bulldog(Dog):
    pass

myDog = Bulldog()
myDog.bark()
```



mung

Class Inheritance (2)



```
class Dog():  
    def __init__(self):  
        print('Dog')  
  
class Bulldog(Dog):  
    pass  
  
myDog = Bulldog()
```



Dog



```
class Dog():  
    def __init__(self):  
        print('Dog')  
  
class Bulldog(Dog):  
    def __init__(self):  
        print('Bulldog')  
  
myDog = Bulldog()
```



Bulldog

Class Inheritance (3)

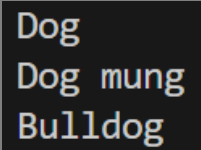


```
class Dog():
    def __init__(self):
        print('Dog')

    def bark(self):
        print('Dog mung')

class Bulldog(Dog):
    def __init__(self):
        super().__init__()
        super().bark()
        print('Bulldog')

myDog = Bulldog()
```



Method Overriding

```
class Dog:
    def bark(self):
        print('mung')

class Bulldog(Dog):
    def bark(self):
        print('kau wa woo')

myDog = Bulldog()
myDog.bark()
```



kau wa woo

```
class Dog:
    def bark(self):
        print('mung')

class Bulldog(Dog):
    def bark(self):
        super().bark()
        print('kau wa woo')

myDog = Bulldog()
myDog.bark()
```



mung
kau wa woo



Python

Advanced level skills

What is **None** in Python ?

- **None** is a special constant in Python that represents the absence of a value or a null value.
- It is an **object** of its own datatype, called `NoneType`.

Example

```
def greet(name=None):  
    if name is None:  
        print("Hello!")  
    else:  
        print(f"Hello, {name}!")  
  
greet()           # Hello!  
greet("Chulsoo") # Hello, Chulsoo!
```

Functions often use **None** as a default to mean “no value provided.”

Annotation in Python

- Annotations are a way to **attach metadata** to variables, function parameters, and return values.
- Most commonly, they are used for **type hints** (to indicate expected types).
- They do not affect runtime behavior (= Annotation **can be omitted**). They're mostly for **readability**.

(1) used as a type hint.

```
def greet(name: str, age: int) -> str:  
    return f"Hello {name}, you are {age} years old."  
  
msg = greet("Chulsoo", 28)  
print(msg)
```

=

```
def greet(name, age):  
    return f"Hello {name}, you are {age} years old."  
  
msg = greet("Chulsoo", 28)  
print(msg)
```

(2) used as a metadata

```
def compute(x: 2, y: 5) -> 10:  
    return x * y  
  
result = compute(5, 10)  
print(result)
```

=

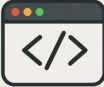
```
def compute(x, y):  
    return x * y  
  
result = compute(5, 10)  
print(result)
```



Iterable objects

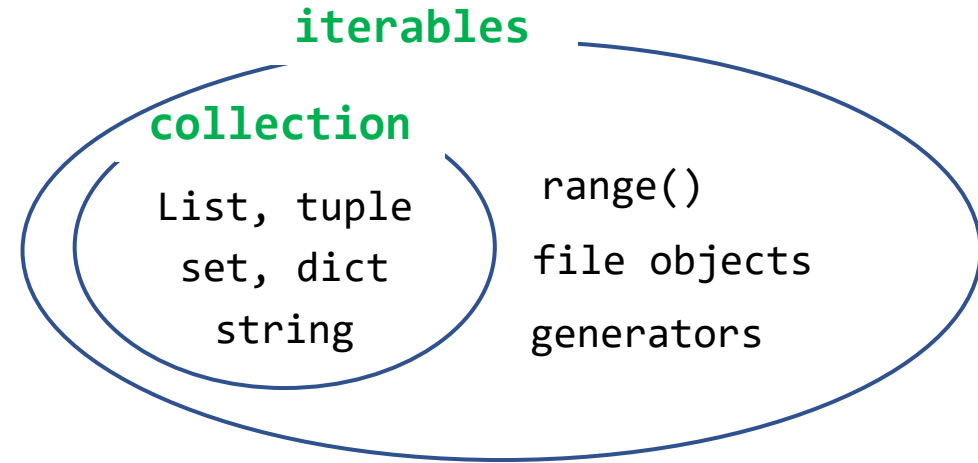


```
for i in [2, -4, 30] :
    print(i)
```



```
2
-4
30
```

for element **in** **iterable**



- An **iterable** object is any object that can be used in a for ... in ... loop.
(= an **object that can be iterated over**).
- All **collections** (list, tuple, set, dict, string) are iterables.
- Other objects like **range()**, **generators**, and **file objects** are also **iterables**.
☞ Therefore, **iterable** $\supset$ **collection** (iterables form a broader concept).



Iterator



```
a = [1, 2, 3]
it = iter(a) # create an iterator from the list
```

```
print(next(it)) # 1
print(next(it)) # 2
print(next(it)) # 3
print(next(it)) # raises StopIteration (no more elements)
```



```
1
2
3
Traceback (most recent call last):
  File "D:\dev\python\myTorch\pytorch\pytorch_test\myTest\simple01.py", line 43, in <module>
    print(next(it))
StopIteration
```

An **iterable** is like a "container" (it can give us an iterator).

An **iterator** is the "machine" that pulls out items from that container one by one.



Custom Iterator

An iterator is an object that implements two special methods:

- `__iter__()` → returns the iterator object itself
- `__next__()` → returns the next value, or raises `StopIteration` when there are no more items



```
class Countdown:
    def __init__(self, start):
        self.current = start

    def __iter__(self):
        return self # the iterator object returns itself

    def __next__(self):
        if self.current <= 0:
            raise StopIteration # no more values
        num = self.current
        self.current -= 1
        return num

for number in Countdown(5): # Using the iterator
    print(number)
```



5
4
3
2
1



Generator

- A generator is a simpler way to **create an iterator** using the **yield** keyword.
- When a function contains **yield**, it automatically becomes a generator function, returning a generator object.
- Generators **automatically implement** `__iter__()` and `__next__()` for us.



```
def countdown(start):  
    while start > 0:  
        yield start # pauses here and returns value  
        start -= 1  
  
# Using the generator  
for num in countdown(5):  
    print(num)
```



5
4
3
2
1



Built-in Functions (1)

enumerate() function



```
animals = ["dog", "cat", "cow"]

# Using enumerate without specifying start
for index, animal in enumerate(animals):
    print(index, animal)

print("-----")

# Using enumerate with a custom start value
for index, animal in enumerate(animals, start=100):
    print(index, animal)
```

```
0 dog
1 cat
2 cow
-----
100 dog
101 cat
102 cow
```

Enumerate() returns **pairs** of (index, element) that we can use in a loop.



Built-in Functions (2)

zip() function (1)

(1) Basic example

```
animals = ["dog", "cat", "cow"]  
ages = [1, 2, 5]
```

```
zipped = zip(animals, ages)  
print(list(zipped))
```

```
[('dog', 1), ('cat', 2), ('cow', 5)]
```



Built-in Functions (3)

zip() function (2)

(2) Iterating with zip()

```
animals = ["dog", "cat", "cow"]
ages = [1, 2, 5]

for animal, age in zip(animals, ages):
    print(f"{animal} is {age} years old.")
```

```
dog is 1 years old.
cat is 2 years old.
cow is 5 years old.
```

(3) Unzipping: reverse of zip()

we can use * to unzip back into separate lists.

```
pairs = [('dog', 1), ('cat', 2), ('cow', 5)]
animals, ages = zip(*pairs)

print(animals)  # ("dog", "cat", "cow")
print(ages)     # (1, 2, 5)
```

```
('dog', 'cat', 'cow')
(1, 2, 5)
```

zip()

```
sentences = ["I love dogs", "Cats are mean", "Dogs are loyal"]  
labels = [1, 0, 1]  
  
pairs = zip(sentences, labels) # creates an iterator that produces these tuples one by one:  
  
print(list(pairs))
```

```
[('I love dogs', 1), ('Cats are mean', 0), ('Dogs are loyal', 1)]
```



Built-in Functions (4)

filter() function

filter(**function**, iterable) applies a **function** to each element of an **iterable** (like a list or tuple)

- It returns only the elements for which the function returns True.
- The result is an iterator (we need to wrap it in list() to see the values).

```
num = [1, 2, 3, 4, 5, 6]
```

```
# extract only even numbers
```

```
def is_even(n):
```

```
    return n % 2 == 0
```

```
result = filter(is_even, num)
```

```
print(list(result))
```

```
[2, 4, 6]
```

Lambda Function (1)

```
# normal function
def add(x, y):
    return x + y

# Equivalent lambda function
add_lambda = lambda x, y: x + y

print(add(3, 5))          # 8
print(add_lambda(3, 5))  # 8
```



8
8

- Lambda functions are good for **short, throwaway functions**.
- They should not be used if the logic is complex (for readability).

Lambda Function (2)

(1) Using Lambda with map()

```
num = [1, 2, 3, 4, 5]

# Square each number
squares = list(map(lambda x: x**2, num))
print(squares)
```



[1, 4, 9, 16, 25]

(2) Using Lambda with filter()

```
num = [10, 15, 20, 25, 30]

# extract only even numbers
evens = list(filter(lambda x: x % 2 == 0, num))
print(evens)
```



[10, 20, 30]



List Comprehension



```
a = [x**2 for x in range(1_000_000)]  
print(a[0:5])
```



```
[0, 1, 4, 9, 16]
```

```
b = [1, 2, 3, 4]  
results = [num*3 for num in b if num%2 == 0]  
print(b)
```



```
[6, 12]
```

List comprehension expression (1)



```
M = [[1,2,3],
      [4,5,6],
      [7,8,9]]
```

```
col_0 = [row[0] for row in M]
```

```
print(col_0) # column 0 of M
print(M)     # unchanged
```



```
[1, 4, 7]
[[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```



```
diag = [M[i][i] for i in range(3)] # range(3) == [0,1,2]
print(diag)
```



```
[1, 5, 9]
```

Generator



```
M = [[1,2,3],  
      [4,5,6],  
      [7,8,9]]
```

```
G = (row[0] for row in M) # generator
```

```
print(G)  
print(next(G))  
print(next(G))  
print(next(G))
```



```
<generator object <genexpr> at 0x00000285580B1490>  
1  
4  
7
```



Generator



```
G = (x * 2 for x in range(3))
print(G)
print(list(G))
```



```
<generator object <genexpr> at 0x00000202D0702500>
[0, 2, 4]
```

In `G = (x * 2 for x in range(3))`, the parenthesis `()` do not create a tuple
– they create a generator expression.

Syntax	Example	Result Type	Characteristics
Tuple	<code>(1, 2, 3)</code>	tuple	Stores all elements in memory at once
Generator expression	<code>(x * 2 for x in range(3))</code>	generator	Produces values one at a time on demand (lazy evaluation)

Tuple: (value1, value2, ..) → a simple comma-separated list of values is a tuple.

Generator: (for .. in ..) → If there is a for loop inside the parenthesis `()`, it's a generator expression.

Generator (1)

- Definition: A **special kind of iterator** that **doesn't store all values in memory at once**.
- Great for memory efficiency, especially when working with very large datasets or potentially infinite sequences.
- **One-time use**: Once a value is consumed, it cannot be retrieved again without recreating the generator.

How to create a generator

(Method 1): Generator Expression

```
G = (x * 2 for x in range(3)) # generator
print(type(G))
print(G)
print(next(G))
print(next(G))
```



```
<class 'generator'>
<generator object <genexpr> at 0x000001D9F98B2500>
0
2
```

(Method 2): Using the **yield** keyword

```
def my_generator():
    yield 1
    yield 2
    yield 3

g = my_generator() # generator
print(next(g)) # 1
print(next(g)) # 2
```



```
1
2
```

Generator (2)

Example: Handling large data



```
def big_numbers():  
    for i in range(1_000_000_000):  
        yield i  
  
g = big_numbers()    # generator  
  
print(next(g))    # 0  
print(next(g))    # 1  
# can generate values indefinitely without memory overflow
```



```
0  
1
```

List vs Generator

List comprehension



```
L = [x**2 for x in range(1_000_000)] # Uses a lot of memory
```

Generator expression



```
G = (x**2 for x in range(1_000_000)) # Uses very little memory
```

A generator is an iterator that produces values **one at a time**, on demand, and **without storing them all in memory**.



Closure (1)

Closure is a **function** that "**remembers**" the variables from the scope in which it was created, even if that scope is no longer active.



```
def outer_function(msg):  
    def inner_function():  
        print("Message:", msg)  
    return inner_function # returning the inner function  
  
# Create closures  
hello_func = outer_function("Hello")  
bye_func = outer_function("Goodbye")  
  
# Even though outer_function has finished execution,  
# the inner functions remember the 'msg' variable  
hello_func() # Output: Message: Hello  
bye_func()   # Output: Message: Goodbye
```



```
Message: Hello  
Message: Goodbye
```



Closure (2)



```
def my_decorator(func):  
    def wrapper():  
        print("Before func()")  
        func()  
        print("After func()")  
        return wrapper # return the closure  
  
def say_hello():  
    print("Hello!")  
  
decorated_func = my_decorator(say_hello)  
  
# decorated_func is actually 'wrapper'  
# which remembers 'func' decorated_func()  
decorated_func()
```



```
Before func()  
Hello!  
After func()
```



Closure (3)



```
def my_decorator(func):  
    def wrapper():  
        print("Before func()")  
        func()  
        print("After func()")  
        return wrapper # return the closure  
  
def say_hello():  
    print("Hello!")  
  
say_hello = my_decorator(say_hello)  
  
# decorated_func is actually 'wrapper'  
# which remembers 'func' decorated_func()  
say_hello()
```



```
Before func()  
Hello!  
After func()
```



Decorator

Decorator is a design pattern that uses closures to modify or extend the behavior of functions.



```
def my_decorator(func):  
    def wrapper():  
        print("Before func()")  
        func()  
        print("After func()")  
    return wrapper
```

```
@my_decorator  
def say_hello():  
    print("Hello!")  
  
say_hello()
```



```
Before func()  
Hello!  
After func()
```

Python

Special Methods (1) (i.e., double underscore methods)

```
class Vector2D:
    def __init__(self, x, y):
        self.x, self.y = x, y

    def __add__(self, other):
        return Vector2D(self.x + other.x, self.y + other.y)

# Vector2D usage
v1, v2 = Vector2D(2, 3), Vector2D(1, 1)

a = v1 + v2
print(a)
print(a.x, a.y)
```

```
<__main__.Vector2D object at 0x000001EB4E80FE50>
3 4
```

c04_special_method1.py

Python

Special Methods (2) : `__repr__()`

```
class Vector2D:
    def __init__(self, x, y):
        self.x, self.y = x, y

    def __add__(self, other):
        return Vector2D(self.x + other.x, self.y + other.y)

    def __repr__(self): # developer-friendly representation
        return f"Vector2D({self.x}, {self.y})"

# Vector2D usage
v1, v2 = Vector2D(2, 3), Vector2D(1, 1)

a = v1 + v2
print(a)
```

Vector2D(3, 4)

c04_special_method1.py

```
class Vector2D:
    def __init__(self, x, y):
        self.x, self.y = x, y

    def __sub__(self, other): # vector subtraction
        return Vector2D(self.x - other.x, self.y - other.y)

    def __mul__(self, scalar): # scalar multiplication
        return Vector2D(self.x * scalar, self.y * scalar)

    def __matmul__(self, other): # dot product
        return self.x * other.x + self.y * other.y

    def __str__(self): # user-friendly string
        return f"({self.x}, {self.y})"

# Vector2D usage
v1, v2 = Vector2D(2, 3), Vector2D(1, 1)

b = v1 - v2
c = v1 * 2 # in this implementation, v1 * v2 causes error
d = v1 @ v2

print(b, c, d)
```

(1, 2) (4, 6) 5

Special Methods (5): `__enter__()` and `__exit__()`

```
class DummyContext:
    # Context manager using __enter__ and __exit__.
    def __enter__(self):
        print("Entering context")
        return "Resource Acquired"

    def __exit__(self, exc_type, exc_val, exc_tb):
        print("Exiting context")

# The standard way to open a file using 'with'
with open('example.txt', 'w') as f:
    f.write("This is a test.")

# Context manager
with DummyContext() as res:
    print(res)
```

```
Entering context
Resource Acquired
Exiting context
```

Objects can be **used in a with block** via `__enter__()` and `__exit__()`.

c04_enter_exit_1.py

- In Python, functions are callable (you can do `f(x)`).
- Objects (instances of classes) are **not** callable by default.
- But if a class defines the special method `__call__`, then **its instances become callable, like functions.**

```
class MyAdder:
    def __init__(self, n):
        self.n = n

    def __call__(self, x):
        return x + self.n

adder = MyAdder(10)  # adder is an instance of MyAdder class
print(adder(5))      # 15 (calls adder.__call__(5))
```

15

- Because we defined `__call__`, we can use `adder(5)` as if `adder` were a function.
- Internally, Python executes `adder.__call__(5)`.

```
class Dog:
    def __call__(self):
        print('mung mung')
```

```
my_dog = Dog()
my_dog()
```

`mung mung`

c04_call1.py

Objects can be called like functions via `__call__()` in the class.

```
class Cat:
    def __call__(self, num):
        print('ya-ong ' * num)
```

```
my_cat = Cat()
my_cat(3)
```

`ya-ong ya-ong ya-ong`

c04_call1.py

- It lets you write objects that **behave like functions**, **but also carry internal state**.
- This is exactly how PyTorch's `nn.Module` works.

- Inside `nn.Module.__call__`, PyTorch does extra work:
 - Calls `.forward()`
 - Applies post-processing

So you can write `criterion(y_hat, y)` instead of `criterion.forward(y_hat, y)`
– much cleaner!

Let `criterion` be an instance of `nn.MSELoss`:

```
criterion = nn.MSELoss().
```

Callable Object via `__call__`

In Python, when we write

```
loss = criterion( $y_{\text{hat}}$ ,  $y$ ),
```

this does not call the constructor again. Instead, it invokes the special method

```
criterion.__call__( $y_{\text{hat}}$ ,  $y$ ).
```

Delegation to forward

For PyTorch modules, the method `__call__` internally delegates to the `forward` function:

```
criterion.__call__( $y_{\text{hat}}$ ,  $y$ )  $\longrightarrow$  criterion.forward( $y_{\text{hat}}$ ,  $y$ ).
```

Mean Squared Error Computation

Specifically, for `nn.MSELoss`, the forward method computes the mean squared error:

$$\mathcal{L}(y_{\text{hat}}, y) = \frac{1}{N} \sum_{i=1}^N (y_{\text{hat},i} - y_i)^2.$$

Summary

Thus,

```
loss = criterion( $y_{\text{hat}}$ ,  $y$ )
```

is equivalent to evaluating the MSE loss between the prediction y_{hat} and the target y .

Thank you!

